
Algorithm 1: L-BFGS Two-Loop Recursion for $r = H_m q$

Input: q , stored pairs $\{(s_i, y_i)\}_{i=0}^{m-1}$, initial matrix H_0

Output: $r = H_m q$

```
1 for  $i = m - 1, m - 2, \dots, 0$  do  
2   |  $\rho_i \leftarrow 1/(y_i^\top s_i)$   
3   |  $\alpha_i \leftarrow \rho_i s_i^\top q$   
4   |  $q \leftarrow q - \alpha_i y_i$   
5 end  
6  $r \leftarrow H_0 q$   
7 for  $i = 0, \dots, m - 1$  do  
8   |  $\beta_i \leftarrow \rho_i y_i^\top r$   
9   |  $r \leftarrow r + s_i(\alpha_i - \beta_i)$   
10 end  
11 return  $r$ 
```
